Housing Lending, Overseas Migration and Wage Growth Across Australian States

Data Exploration Project

TingTing Wu
Student ID: 35717343

FIT5147 Data Exploration and Visualization
Semester 1, 2026
Monash University

29 March 2026

Contents

Table of contents

1 Introduction **3**

2 Data Wrangling and Checking **3**
- 2.1 Creating a Common Financial Year Structure . . . 4
- 2.2 Understanding Missing Values in the Housing Finance Data . . . 4
- 2.3 Checking Distribution Shapes Across the Datasets . . . 5
- 2.4 Summary of Data Checking . . . 5

3 Data Exploration **5**
- 3.1 Housing Lending, Wage Growth and Lending Structure Across Australian States . . . 6
  - 3.1.1 Divergence between housing lending growth and wage growth . . . 6
  - 3.1.2 Long-run changes in the investor share of lending value . . . 7
  - 3.1.3 Summary of findings . . . 7
- 3.2 Changing Migrant Origins, State Settlement Patterns, and Their Links to Wage Growth and Housing Lending in Australia . . . 8
  - 3.2.1 Changing Origins of Migrant Inflows Across Australian States . . . 8
  - 3.2.2 Within-state relationships between migrant-origin concentration and economic outcomes . . . 9
  - 3.2.3 Summary of findings . . . 10
- 3.3 Where are the largest gaps between investor lending and migration demand across Australian states, and how has this geography changed over time? . . . 10

4 Conclusion **12**

5 Reflection **12**

6 References **12**

A Appendix **12**
- A.1 Key calculated values for Housing Lending, Wage Growth and Lending Structure Across Australian States . . . 12
  - A.1.1 Main values cited in the annual growth discussion . . . 12
  - A.1.2 Main values cited in the investor-share discussion . . . 12
- A.2 Further calculation details for Within-State Links Between Migrant-Origin Concentration and Economic Outcomes . . . 13
  - A.2.1 Main values cited in the origin-composition discussion . . . 14
  - A.2.2 Main values cited in the robust within-state regression discussion . . . 14

1 Introduction

This report, *Housing Lending, Overseas Migration and Wage Growth Across Australian States*, examines housing affordability in Australia, a major social and economic issue. Public debate often links housing pressure to migration, limited supply, and weak wage growth. For example, SBS News (2026) discussed whether high migration is truly supported by evidence as a cause of the housing crisis. This makes the topic important for households, workers, students, and policy makers.

The report explores how housing lending, overseas migration, and wage growth have changed across Australian states over time. Using Australian Bureau of Statistics data, it examines owner-occupier lending, investor lending, net overseas migration, and the Wage Price Index. The aim is to assess whether common public claims about migration, wages, and housing stress are reflected in the data, and whether these patterns differ by state.

I am personally interested in this topic because, as an international student from Taiwan living in Melbourne, I have become more aware of housing affordability pressures. Seeing how difficult high housing costs can make financial security for students and workers motivated me to examine this issue using data rather than relying only on public opinion or media debate.

This report addresses the following questions:

1. Does the growth of housing lending align with wage growth across Australian states, and how has the lending structure changed over time?
2. How have the origins of overseas migrants and their state settlement patterns changed across Australian states over time?
3. Where are the largest gaps between investor lending and migration demand across Australian states, and how have these gaps changed over time?

2 Data Wrangling and Checking

This project uses four open datasets from the Australian Bureau of Statistics (ABS): Net Overseas Migration by Country of Birth, Owner-Occupier Lending, Investor Lending, and the Wage Price Index. The direct download links are provided below:

- Net Overseas Migration by Country of Birth
- Owner-Occupier Lending
- Investor Lending
- Wage Price Index

Before checking and cleaning the data, I first transformed the three ABS datasets into tidy formats that were easier to compare and analyse in R. Although all three sources were downloaded as Excel workbooks, they had different internal structures and could not be used directly for analysis.

For the overseas migration data, the workbook contained nine separate tables, including one national table and eight state or territory tables. I created a lookup table containing the sheet names, state names, and row ranges, then used a function to read each worksheet in the same way. In each sheet, I used the first imported row as the column names, renamed the first two columns as `sacc_code` and `country_of_birth`, and removed note labels from the year columns, which is the letter attached to 2024-25. I then converted the data from wide format to long format using `pivot_longer()`, so that each row represented one state, one country of birth, and one financial year. Finally, I combined all nine sheets into a single dataset called `overseas_migration`.

For the housing finance data, I used the `Data1` sheet from both the owner-occupier and investor lending workbooks. In both files, the first ten rows contained metadata rather than observations, so I read these rows separately from the main time series data. I created temporary column names for the series columns and extracted key metadata such as state, value type, unit, and series type by splitting the original ABS series description. I then reshaped the time series data into long format

and created additional time variables, including calendar year, quarter, and financial year. After that, I combined the owner-occupier and investor datasets into one dataset called `housing_finance`, with an added variable called `loan_type` to distinguish between the two sources.

For the Wage Price Index data, I again used the `Data1` sheet and followed a similar process. I read the metadata rows separately, created temporary series column names, and extracted information such as state, wage measure, and unit from the ABS series descriptions. I then reshaped the data into long format and created year, quarter, and financial year variables. This produced a final tidy dataset called `wage_price_index`.

Overall, the main purpose of this step was to convert the original Excel workbooks into consistent long-format datasets that could be compared more easily across states and time. This also made it easier to carry out the next stage of the project, which involved checking the structure, identifying missing values, reviewing inconsistencies, and cleaning the data where necessary. The tidy datasets and all data processing code used in this project are available in my GitHub repository.

2.1 Creating a Common Financial Year Structure

As shown in Table 1, Housing finance and Wage price index are quarterly datasets recorded using calendar year and month, while Overseas migration is recorded by financial year. To make the three datasets easier to compare, I created a financial year variable for the quarterly datasets based on the quarter-ending months of September, December, March, and June. The datasets also have different time ranges, since Housing finance and Wage price index start earlier than Overseas migration. Therefore, I restricted all three datasets to the shared financial-year period from 2004-05 to 2024-25 before carrying out the analysis.

Table 1: Summary of time structure across the datasets

Dataset	Start (Year/Month)	End (Year/Month)	Frequency	Time period
Overseas migration	Not applicable	Not applicable	Annual	2004-05 to 2024-25
Housing finance	2002/9	2025/12	Quarterly	2002-03 to 2025-26
Wage price index	1997/9	2025/12	Quarterly	1997-98 to 2025-26

2.2 Understanding Missing Values in the Housing Finance Data

Table 2 shows that the housing finance dataset is largely complete, except for the Investor Number series, which is only available from 2019-20 onward. This suggests that the missing values are structural rather than accidental, since the series begins later in the original source data. Therefore, these values were not imputed, and any analysis using Investor Number should be limited to 2019-20 onward or interpreted cautiously when compared with the longer series.

Table 2: Series start and end dates in the original housing finance workbooks

Loan type	Value type	First FY	Last FY	Missing values	Available values
Investor	Number	2019-20	2024-25	1620	648
Investor	Value	2004-05	2024-25	0	2268
Owner-occupier	Number	2004-05	2024-25	0	2268
Owner-occupier	Value	2004-05	2024-25	0	2268

2.3 Checking Distribution Shapes Across the Datasets

Figure 1 shows no clear sign of major data errors after the asinh transformation. The housing finance data are right-skewed, but the distribution is still fairly smooth, so this looks more like normal variation than unusual mistakes. The overseas migration data are strongly concentrated around zero, with longer tails on both sides, which suggests that many observations are small or zero while some are much larger in either the positive or negative direction. The wage price index data show a clear bimodal shape, but this is likely due to combining index values and percentage change measures in the same dataset, rather than indicating abnormal observations. Overall, this check suggests that the main issue is differences in data structure across the datasets, not obvious errors in the values.

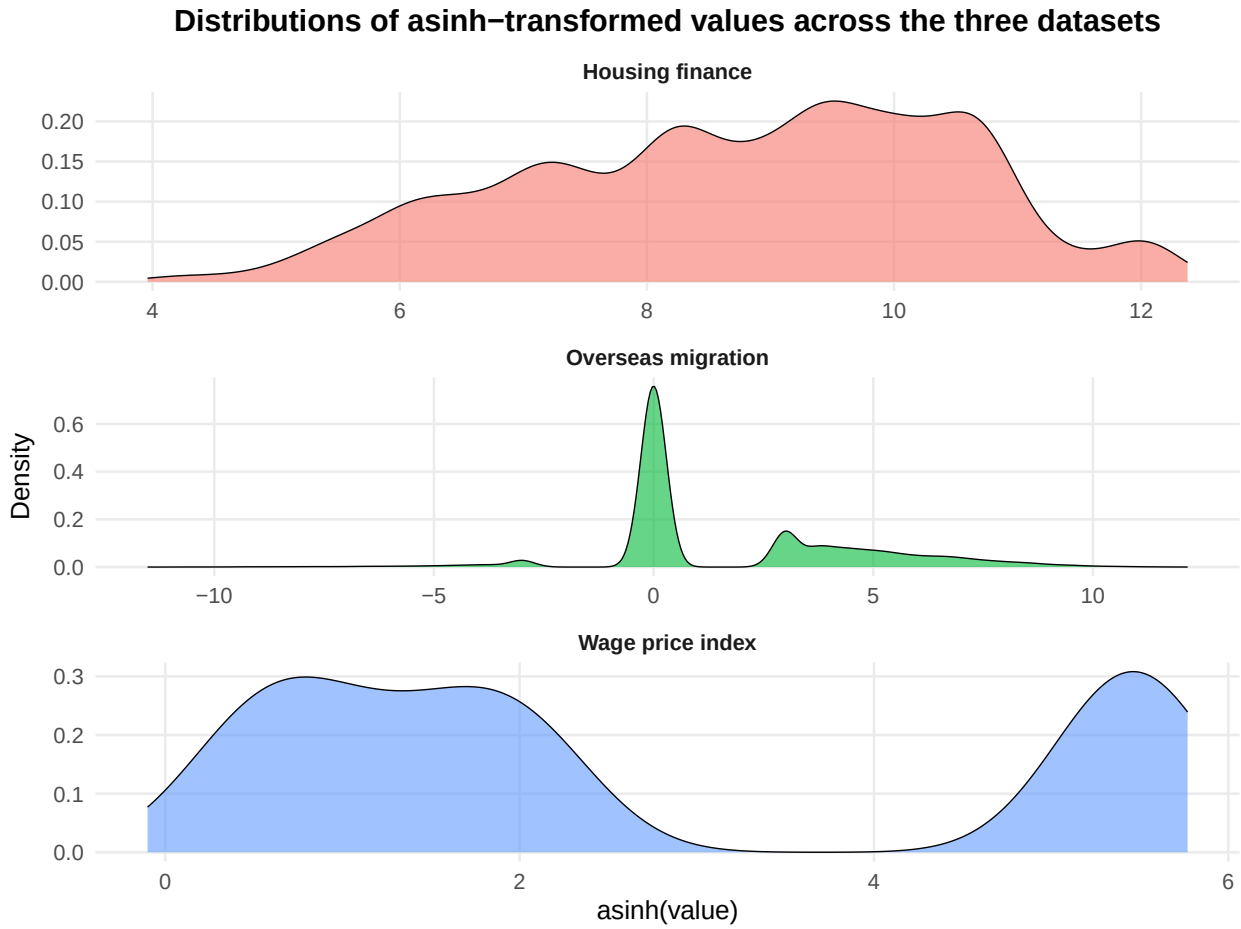

Figure 1

2.4 Summary of Data Checking

Overall, the data wrangling and checking steps suggest that the three datasets were successfully converted into tidy formats and aligned to a common financial-year structure for comparison. No major data quality problems were identified, aside from the structurally missing Investor Number series and the expected differences in measurement scales across the datasets. These issues were taken into account in later analysis, so the cleaned data provide a suitable basis for exploring relationships between migration, housing lending, and wage growth across Australian states.

3 Data Exploration

This section explores the **relationships between housing lending, overseas migration, and wage growth across Australian states** using financial-year summaries and composition measures

from **2004-05 to 2024-25**.

3.1 Housing Lending, Wage Growth and Lending Structure Across Australian States

To address the first research question, I compared annual housing lending growth with annual wage growth across Australian states from 2005-06 to 2024-25. I then examined whether these two measures moved together over time, and how the investor share of total lending value changed across states.

3.1.1 Divergence between housing lending growth and wage growth

Figure 2 separates housing lending growth and wage growth into two panels, making their state-level patterns easier to compare. Wage growth stays within a relatively narrow range of about 1.5% to 4.2%, but still shows a longer-run decline from 2005-06 to 2020-21 before rising again towards its period average of 3%. Western Australia shows the largest swings in wage growth, while Tasmania appears least affected during the COVID-19 period and rebounds strongly afterwards.

By contrast, housing lending growth is much more volatile and differs more across states. The sharp divergence in 2020-21 likely reflects the COVID-19 period, while the large fall in 2022-23 is consistent with a post-pandemic adjustment. Lending fluctuations are especially large in Northern Territory, Western Australia, and New South Wales, whereas South Australia and the Australian Capital Territory are relatively more stable. Overall, the figure suggests that short-run changes in housing finance are driven much more by lending market conditions than by wage growth.

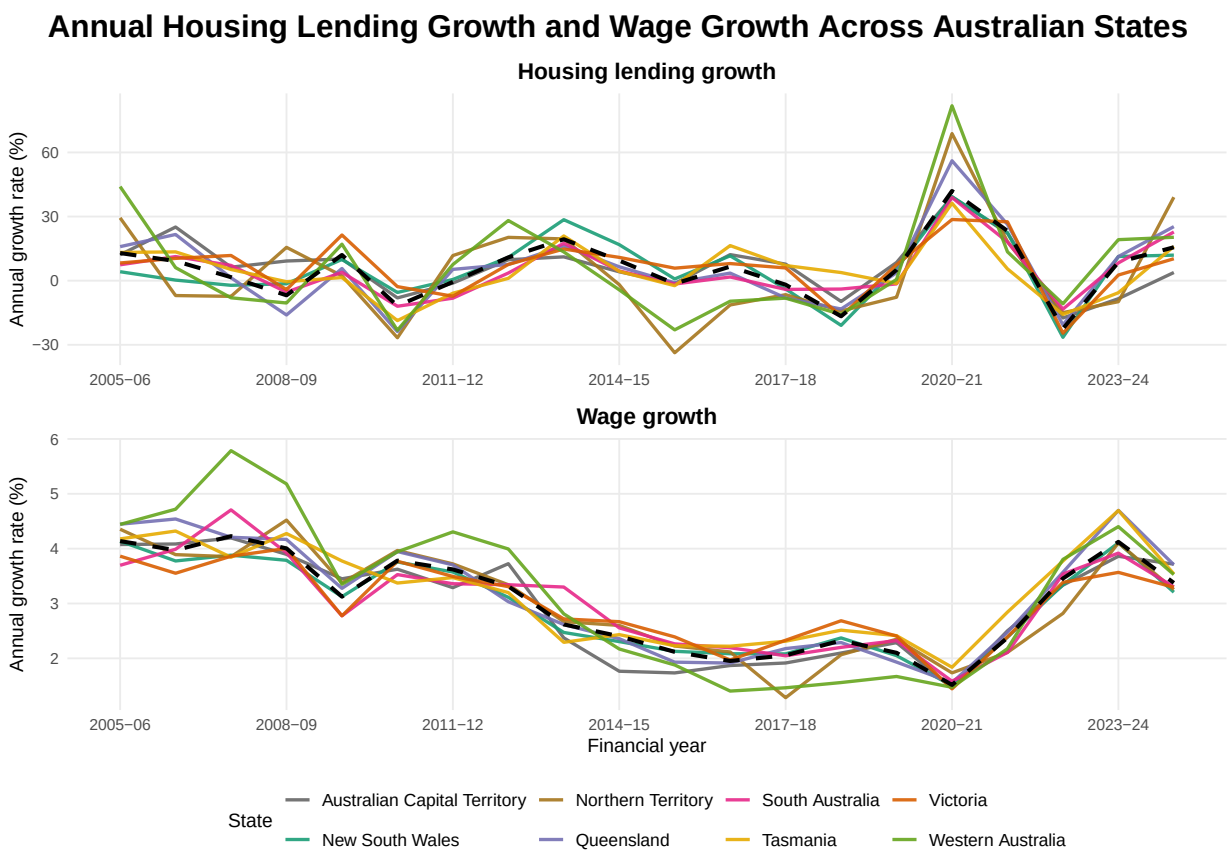

Figure 2: Annual housing lending growth and wage growth across Australian states, 2005-06 to 2024-25. Coloured lines show individual states and the dashed black line shows the Total Australia series.

3.1.2 Long-run changes in the investor share of lending value

Figure 3 examines how the investor share of total housing lending value changed across states from 2004-05 to 2024-25. The national pattern is clearly cyclical. The investor share starts at about 38.3%, rises to a peak of 44.5% in 2014-15, then falls sharply to 25.4% in 2020-21 before recovering to 37.8% by 2024-25.

New South Wales stands out as the most investor-oriented market over the long run. Its average investor share is about 39.6%, and it remains above the national share in 81% of years, peaking at 53.9% in 2014-15. By contrast, Tasmania has one of the lowest average investor shares, at about 25.8%. Northern Territory and Tasmania show a somewhat similar long-run pattern, with investor shares generally remaining below the national benchmark for much of the period. However, both rise sharply after the COVID-19 period. By 2024-25, Northern Territory reaches 40.5% and Tasmania reaches 26.2%, compared with the national share of 37.8%. Western Australia follows a different pattern: after 2008-09, its investor share begins to move further away from the national benchmark, before recovering again after the 2020-21 low. It rises from 16.1% in 2020-21 to 36.8% in 2024-25. The other states generally follow the national cycle more closely and do not show such distinctive departures. Overall, the figure suggests that the structure of housing lending has changed substantially over time and that investor activity has been much more important in some state housing markets than in others.

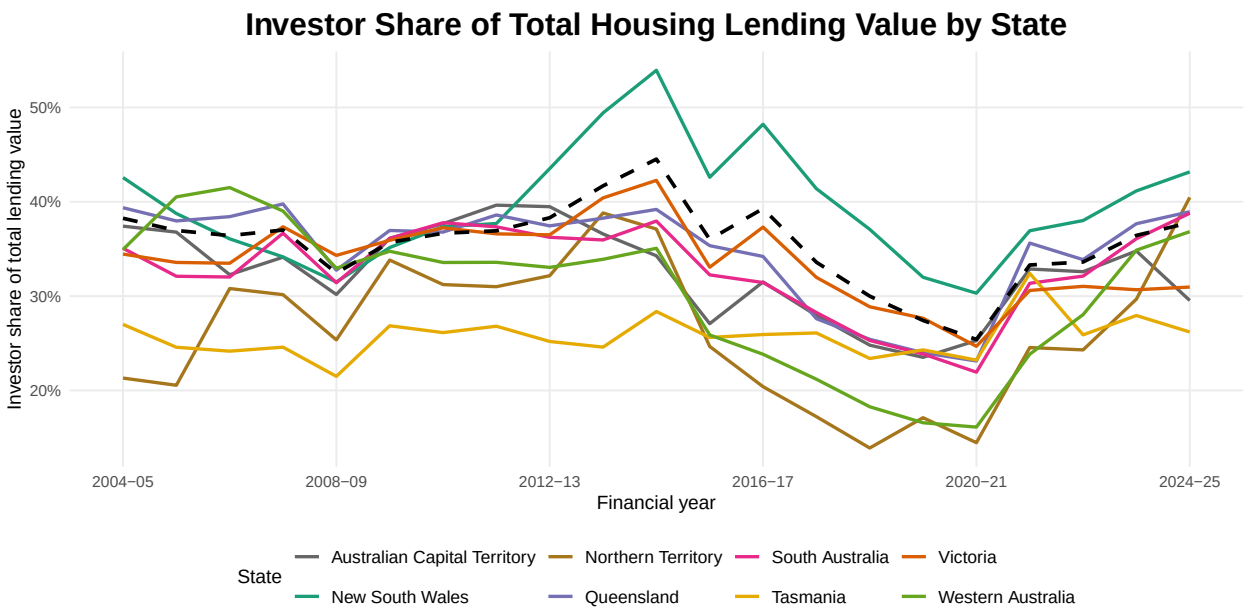

Figure 3: Investor share of total housing lending value across Australian states, 2004-05 to 2024-25. Coloured lines show individual states and the dashed black line shows the Total Australia share.

3.1.3 Summary of findings

Overall, Figure 2 and Figure 3 suggest that housing lending growth does not closely track wage growth across Australian states, and that the structure of lending has changed markedly over time. Investor activity appears to be more strongly related to the composition of lending than to short-run movements in total lending growth, with some states remaining much more investor-oriented than others.

3.2 Changing Migrant Origins, State Settlement Patterns, and Their Links to Wage Growth and Housing Lending in Australia

To address the second research question, I examined how the origins of overseas migrants changed across Australian states from 2004-05 to 2024-25. I then examined whether migrant-origin concentration within each state was associated with wage growth and housing lending growth.

3.2.1 Changing Origins of Migrant Inflows Across Australian States

Figure 4 shows a clear long-run shift in the origins of migrant inflows across most states. The most striking pattern is the rising importance of Southern and Central Asia, whose national share of positive overseas migrant inflows increases from about 15.6% in 2004-05 to 36.3% in 2024-25. This rise appears across nearly all states, but is especially strong in the Australian Capital Territory and Victoria, where Southern and Central Asia accounts for about 55.6% and 41.0% of positive inflows respectively in 2024-25.

By contrast, North-West Europe becomes less important in most states, although it remains more prominent in Western Australia, where it still accounts for about 21.0% of positive inflows in 2024-25. Overall, migrant settlement patterns remain uneven across states, with Victoria and the Australian Capital Territory becoming more concentrated in Southern and Central Asian inflows, while Western Australia retains a more visible North-West European component.

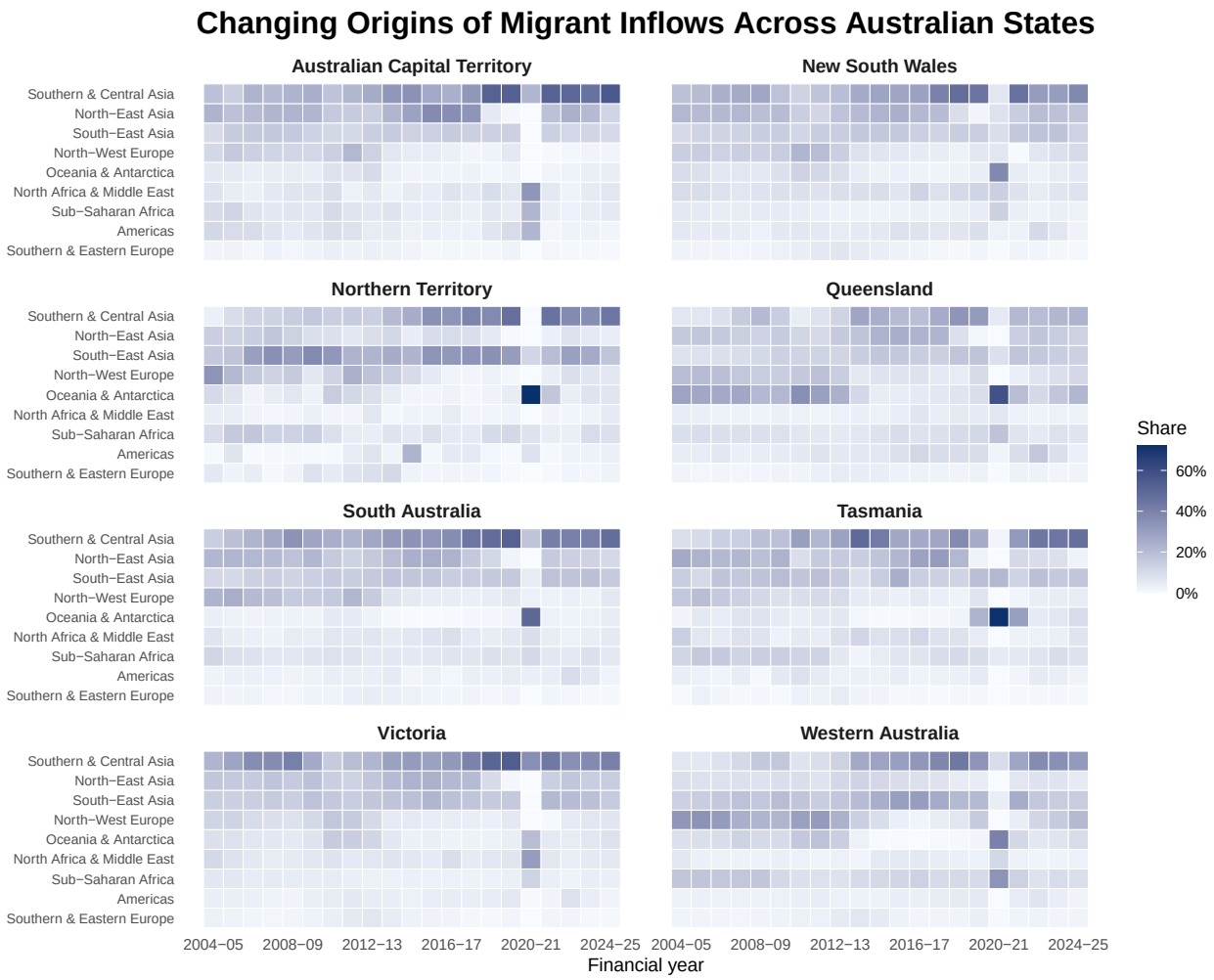

Figure 4: Origin composition of positive overseas migrant inflows across Australian states, 2004-05 to 2024-25. Darker tiles indicate larger shares within each state-year.

3.2.2 Within-state relationships between migrant-origin concentration and economic outcomes

To summarise the relationship more clearly, I estimated a separate within-state slope for each state rather than plotting all state-years directly. Migrant-origin concentration was measured using a Herfindahl index (HHI), where higher values indicate that migrant inflows were concentrated in fewer countries of birth. I then examined whether years with relatively higher concentration within each state also tended to have relatively higher wage growth or housing lending growth. To reduce the influence of unusual observations, I used robust regression and plotted the resulting coefficients with 95% confidence intervals in Figure 5. Further calculation details for this within-state analysis are provided in the Appendix.

In Figure 5, positive coefficients indicate that, within a state, years with more concentrated migrant inflows tended to coincide with stronger-than-usual economic growth, while negative coefficients indicate the opposite. In the wage-growth panel, 7 of the 8 states have negative coefficients, with the strongest negative slopes appearing in New South Wales (-29.5) and the Australian Capital Territory (-23.5). The confidence intervals for New South Wales, Victoria, Tasmania, and the Australian Capital Territory remain below zero, suggesting that higher migrant-origin concentration was associated with weaker-than-usual wage growth in these states.

By contrast, the housing-lending panel is more mixed, with 5 of the 8 states have positive coefficients, but only Northern Territory and Tasmania have intervals that remain clearly above zero, with estimated slopes of about 141.1 and 273.5 respectively. The remaining states have wide intervals that cross zero. Overall, these results suggest that while migrant origins and settlement patterns changed substantially across states, higher migrant-origin concentration was more consistently linked to weaker wage growth than to housing lending growth, and its relationship with housing lending remained uneven across Australia.

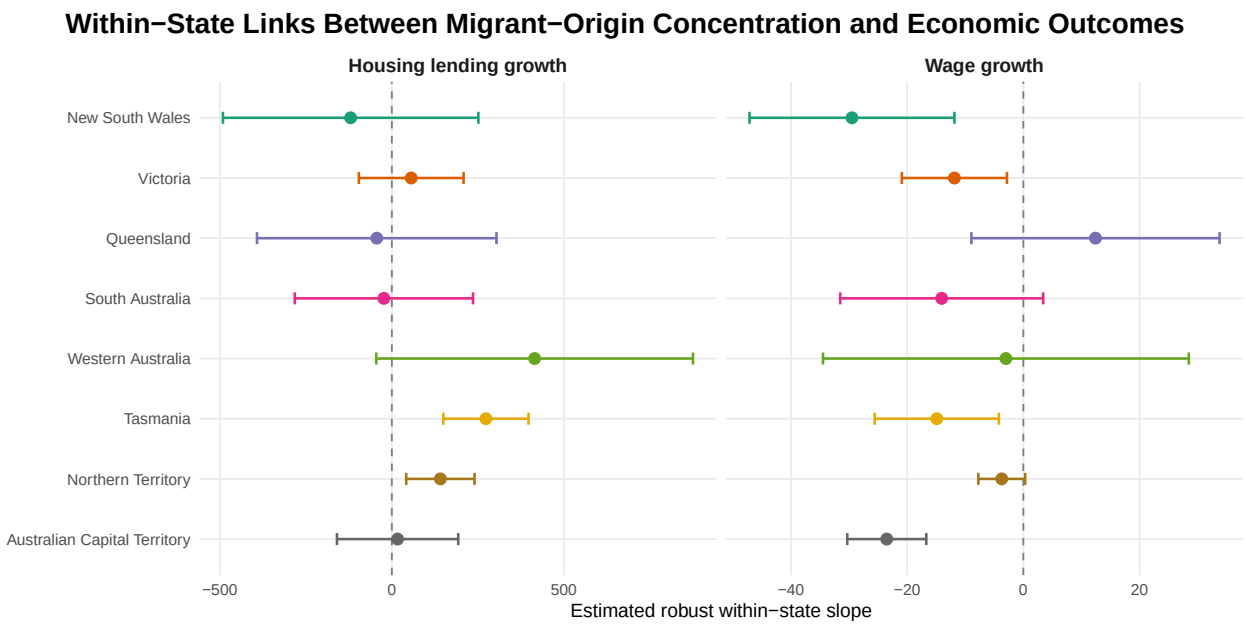

Figure 5: Robust within-state associations between migrant-origin concentration and economic outcomes, 2005-06 to 2024-25. Coefficients are estimated from state-specific robust regressions using deviations from each state's own average level. Positive coefficients indicate that years with above-average migrant-origin concentration tend to coincide with above-average economic growth within the same state.

3.2.3 Summary of findings

Overall, Figure 4 and Figure 5 suggest that the origins of overseas migrants and their settlement patterns changed substantially across Australian states over time. Migrant inflows became increasingly dominated by Southern and Central Asia, but this shift was not uniform across Australia, with Victoria and the Australian Capital Territory showing stronger concentration in Southern and Central Asian inflows, while Western Australia retained a more visible North-West European component. However, these changes in migrant origins were not associated with stronger economic outcomes in any consistent way. Within states, higher migrant-origin concentration was more often linked to weaker wage growth, while its relationship with housing lending growth was mixed and uneven across Australia.

3.3 Where are the largest gaps between investor lending and migration demand across Australian states, and how has this geography changed over time?

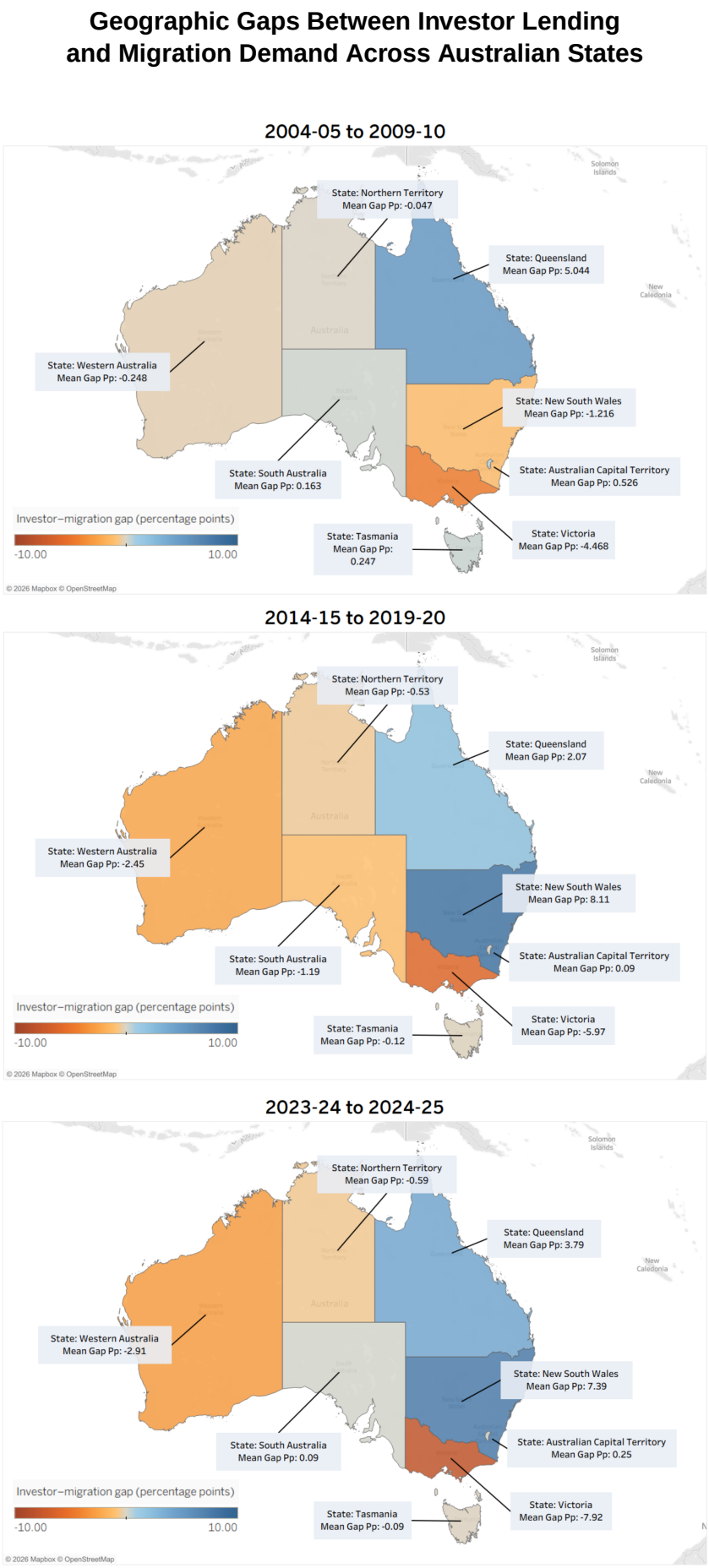

Figure 6: Average gaps between investor lending share and migration-demand share across Australian states in three representative periods. Positive values indicate that a state's share of investor lending exceeded its share of migration demand, while negative values indicate the opposite.

4 Conclusion

5 Reflection

6 References

A Appendix

A.1 Key calculated values for Housing Lending, Wage Growth and Lending Structure Across Australian States

This appendix reports the main calculated values cited in Questions 1 and 2.

For Question 1, annual growth rates were calculated as:

$$Growth_{st} = \left(\frac{X_{st}}{X_{s,t-1}} - 1 \right) \times 100$$

where X_{st} is the annual value for state s in financial year t.

The investor share of total housing lending value was calculated as:

$$InvestorShare_{st} = \frac{InvestorValue_{st}}{InvestorValue_{st} + OwnerOccupierValue_{st}}$$

The proportion of years in which a state's investor share exceeded the national share was calculated as:

$$AboveNational_s = \frac{1}{T} \sum_t I\left(Share_{st} > Share_t^{AUS}\right)$$

where $I(\cdot)$ is an indicator that takes the value 1 when the condition is true and 0 otherwise.

A.1.1 Main values cited in the annual growth discussion

Table 3: Main values cited in the discussion of annual housing lending growth and wage growth

Measure	Value
National wage growth minimum	1.5%
National wage growth maximum	4.2%
National average wage growth	3%

A.1.2 Main values cited in the investor-share discussion

Table 4: Main values cited in the discussion of long-run changes in the investor share of lending value

Measure	Geography	Financial year	Value
National investor share at start of period	Australia	2004-05	38.3%
National investor share at peak	Australia	2014-15	44.5%

National investor share at trough	Australia	2020-21	25.4%
National investor share in latest year	Australia	2024-25	37.8%
New South Wales average investor share	New South Wales	2004-05 to 2024-25	39.6%
New South Wales share above national benchmark	New South Wales	2004-05 to 2024-25	81.0%
New South Wales peak investor share	New South Wales	2014-15	53.9%
Tasmania average investor share	Tasmania	2004-05 to 2024-25	25.8%
Northern Territory investor share in latest year	Northern Territory	2024-25	40.5%
Tasmania investor share in latest year	Tasmania	2024-25	26.2%
Western Australia investor share in 2020-21	Western Australia	2020-21	16.1%
Western Australia investor share in latest year	Western Australia	2024-25	36.8%

A.2 Further calculation details for Within-State Links Between Migrant-Origin Concentration and Economic Outcomes

This appendix provides further details of the calculation used in the figure titled *Within-State Links Between Migrant-Origin Concentration and Economic Outcomes.*

For each state s and financial year t, migrant-origin concentration was measured using a country-level Herfindahl index (HHI):

$$HHI_{st} = \sum_i p_{ist}^2$$

where p_{ist} is the share of positive net overseas migration from country of birth i in state s and year t. It was calculated as:

$$p_{ist} = \frac{\max(\text{NOM}_{ist}, 0)}{\sum_j \max(\text{NOM}_{jst}, 0)}$$

A higher HHI indicates that migrant inflows were more concentrated in fewer countries of birth, while a lower HHI indicates a more diverse origin mix.

To focus on within-state variation rather than long-run differences between states, migrant-origin concentration, wage growth, and housing lending growth were centred around each state's own average level:

$$HHI_{st}^{dev} = HHI_{st} - \bar{HHI}_s$$

$$WAGE_{st}^{dev} = WAGE_{st} - \overline{WAGE}_s$$

$$LEND_{st}^{dev} = LEND_{st} - \overline{LEND}_s$$

where $\bar{HHI}_s$, $\overline{WAGE}_s$, and $\overline{LEND}_s$ are the long-run state averages for each state.

Separate robust regression models were then estimated for each state. For wage growth, the model was:

$$WAGE_{st}^{dev} = \alpha_s + \beta_s HHI_{st}^{dev} + \varepsilon_{st}$$

For housing lending growth, the model was:

$$LEND_{st}^{dev} = \alpha_s + \beta_s HHI_{st}^{dev} + \varepsilon_{st}$$

The coefficient β_s measures whether years with above-average migrant-origin concentration within a state also tended to have above-average wage growth or housing lending growth in that same state. Robust regression was used so that unusual observations had less influence on the estimated slopes than in ordinary least squares. The intervals shown in the figure are approximate 95% confidence intervals based on the robust standard errors of the fitted models.

A.2.1 Main values cited in the origin-composition discussion

Table 5: Main values cited in the discussion of changing migrant origins across Australian states

Measure	Geography	Financial year	Value
Southern and Central Asia share of positive overseas migrant inflows	Australia	2004-05	15.6%
Southern and Central Asia share of positive overseas migrant inflows	Australia	2024-25	36.3%
Southern and Central Asia share of positive inflows	Australian Capital Territory	2024-25	55.6%
Southern and Central Asia share of positive inflows	Victoria	2024-25	41.0%
North-West Europe share of positive inflows	Western Australia	2024-25	21.0%

A.2.2 Main values cited in the robust within-state regression discussion

Table 6: Main robust regression coefficients and confidence intervals cited in the within-state analysis of migrant-origin concentration and economic outcomes

State	Economic outcome	Estimate	95% CI
Australian Capital Territory	Wage growth	-23.5	[-30.3, -16.7]
New South Wales	Wage growth	-29.5	[-47.1, -11.9]
Tasmania	Wage growth	-14.9	[-25.6, -4.2]
Victoria	Wage growth	-11.9	[-20.9, -2.8]
Northern Territory	Housing lending growth	141.1	[41.9, 240.4]
Tasmania	Housing lending growth	273.5	[149.7, 397.3]